

Hjalmar Fors

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EDUCATION

Carnegie Mellon University — *M.S. in Machine Learning* **2019 - 2021**

- Thesis: 'Efficient Fine-tuning Strategies for Large Language Models in Low-Resource Domains' published at ACL 2021 with 287 citations.

University of Washington — *B.S. in Computer Science* **2015 - 2019**

- Cum laude, 3.87 GPA; published undergraduate research on neural architecture search achieving 89% ImageNet accuracy with 60% fewer FLOPs than baseline.

EXPERIENCE

Anthropic — *Research Engineer - Foundation Models* **June 2023 - Present**

- Optimized multi-head attention mechanisms reducing model inference latency by 18% while maintaining performance parity on MMLU and HumanEval benchmarks.
- Implemented constitutional AI feedback collection pipeline processing 2.3M preference-based training samples, improving RLHF convergence time by 35%.
- Led experimental design and execution for 7 prompt engineering studies across 12 model variants, generating 3 internal research publications on in-context learning phenomena.

DeepMind (Alphabet) — *Machine Learning Engineer - AI Safety* **January 2022 - May 2023**

- Developed interpretability analysis framework using sparse autoencoders to decompose 1024-dimensional activation spaces, identifying 156 distinct behavioral features in language models.
- Designed and executed adversarial robustness evaluation suite for 8 model architectures, discovering critical failure modes affecting 4.2% of safety-critical prompts.
- Collaborated with 4-person team to scale model mechanistic interpretability research from toy models to 7B parameter transformers, improving intervention success rate from 23% to 68%.

OpenAI — *Junior AI Researcher* **August 2021 - December 2021**

- Implemented and benchmarked 5 competing approaches to neural scaling law extrapolation, achieving 12% lower prediction error on held-out 40B model training curves.
- Accelerated data preprocessing pipeline for vision-language models by 6x through custom CUDA kernels, reducing per-epoch compute cost from 8 hours to 80 minutes on A100 clusters.
- Contributed novel regularization technique increasing zero-shot transfer performance by 7.3 points on downstream CV benchmarks across 3 model families.

PROJECTS

Token Prediction Efficiency Study — *PyTorch, FSDP, Weights & Biases, NVIDIA A100s* **March 2024 - Present**

- Engineering efficiency comparative analysis across 11 different token prediction methods on 13B model scale, targeting 25% latency reduction for serving.
- Implemented dynamic batching scheduler improving GPU utilization from 67% to 92% under real-world request distributions from 15 concurrent users.

Constitutional AI Alignment Benchmark — *JAX, Ray Tune, Custom evaluation harnesses* **September 2023 - February 2024**

- Co-authored reproducible benchmark suite evaluating alignment of 8 open-source models against 42 constitutional principles, cited by 12 downstream papers.
- Designed scoring rubric achieving 94% inter-annotator agreement across 5,200 evaluation instances, enabling reliable comparative analysis of model behaviors.

Mechanistic Interpretability Toolkit — *PyTorch, ONNX, FastAPI, Docker* **June 2022 - August 2023**

- Developed open-source library enabling circuit analysis on transformer models, achieving 8.5K GitHub stars and adoption by 6 major AI safety research labs.
- Optimized computation graph for feature attribution extraction reducing 48-layer model analysis time from 47 minutes to 6 minutes via vectorized operations.

EXTRACURRICULAR

NeurIPS Conference — *Program Committee Member* **2023 - Present**

- Reviewed and provided detailed feedback on 18 papers in foundation models and interpretability tracks, contributing to acceptance decisions for top 23% of 9,400 submissions.
- Mentored 4 junior researchers in paper submission strategy and experimental rigor, resulting in 3 papers achieving poster presentations.

- Delivered 6-lecture series on neural scaling laws and interpretability techniques to 140 graduate students, receiving 4.7/5.0 teaching evaluation scores.
 - Designed hands-on PyTorch exercises adopted into curriculum for 3 subsequent semesters across 420+ enrolled students.
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TECHNICAL SKILLS

Languages – Python, C++, MATLAB, Bash

Technologies – PyTorch, TensorFlow, JAX, Hugging Face Transformers, ONNX, Weights & Biases, Ray Tune, Triton, CUDA

Tools – Git, Docker, Kubernetes, Slurm, Weights & Biases, Neptune, Jupyter, MLflow, Hydra